

Małgorzata Nowicka

Doctoral candidate in Visual Communication Design
School of Arts, Design, and Architecture, Aalto University
malgorzata.nowicka@aalto.fi
<https://margonowicka.com>
 <https://orcid.org/0009-0009-2531-9820>

1 Degrees

2023 — Master of Arts (Art and Design) in New Media Design and Production. Aalto University, Espoo, Finland.
Final grade average: 4.87/5 (Excellent). Thesis repository: <https://urn.fi/URN:NBN:fi:aalto-202312037039>

2021 — Bachelor of Science (Engineering) in Computer Science. Warsaw University of Technology, Warsaw, Poland.
Final grade average: 4.71/5 (Excellent).

2 Other education and expertise

2020 — Student exchange in Information and Communication Technology. KTH Royal Institute of Technology, Stockholm, Sweden.

3 Language skills

Polish: native
English: C1 (fluent)

4 Current employment

8.2025-present — Doctoral candidate in Visual Communication Design, supervised by professor Masood Masoodian, in the Aalto Visual Communication Design research group, at the research career stage: I. Part-time (50%). Aalto University, Espoo, Finland.

5.2025-12.2026 — Researcher (CHRYSES: Mapping Environmental Health Crises — Public Understanding through Myths and Science) and animator (“INT-ACT: Intangible Cultural Heritage, Bridging the Past, Present and Future). Part-time (50%). Aalto University, Espoo, Finland.

5 Previous work experience

2.-10.2024 — “Lines of Descent” exhibition designer.
Graphic design, curation, and technical setup for the exhibition at Helsinki Central Library Oodi on the collaboration between the Information Design focus group at Aalto University and FinnGen genetic research project. Part-time (50%).
Aalto University, Espoo, Finland.

3.-8.2023 — MA thesis production.
Directed and animated the 2D short animation “Aurinko and Selene” as a visualization of Finnish genetic phenomenon. Full-time.
Aalto University in collaboration with FinnGen, Helsinki, Finland.

2.2025;

2.,9.2024;

2.-5.2023 — Projection Designer in the “Rottien pyhimys” musical.

Creative and technology designer for a musical directed by Reetta Ristimäki based on the novel by Anneli Kanto. Developing queueing software in TouchDesigner, designing communication protocol between operating computers, setting up the hardware, producing the video content. Freelance and project-based (variable hours).

Greta Tuotanto, Helsinki, Finland.

1.-3.2023 — Generative Video Designer in the “Ihmisen jälkeen / Posthuman” chamber opera.

Video production for an experimental chamber opera written and directed by Temu Mäki in collaboration with composer Max Savikangas. Developed generative and artificial life simulations, 3D animation. Project-based (variable hours).

Greta Tuotanto, Helsinki, Finland.

10.-12.2022 — Lecturer and Teaching Assistant in Aalto University.

Teaching of an introductory course on web development for visual communication Master’s students and assistance in the teaching of a creative coding course. Part-time (50%).

Aalto University, Espoo, Finland.

6.-8.2022 — Research Assistant in a science communication project in Aalto University.

Visualization of the human genome analysis data for a scrollytelling website as a part of the genetic research project FinnGen. Full-time summer internship.

Aalto University in collaboration with FinnGen, Helsinki, Finland.

7.-9.2020;

7.2021-5.2022 — Research Assistant in Natural Computation research group in Aalto University.

Scientific visualization of the interactions between DNA strands (DNA strand displacement) utilizing algorithmic and graph-theoretic approach. Two full-time summer internships and part-time (20%) work on the conference article and posters.

Aalto University, Espoo, Finland.

6 Research and artistic output

6.1 Publications

2025-present — Production of short 2D animations for the “INT-ACT” project as part of the exhibitions about the intangible cultural heritage in Scotland, Finland and Greece.

To be presented at EASA 2026 Anthropology: Possibilities in a Polarised World conference.

2022 — Poster presentation of the DNA visualization software in:

International Conference on Unconventional Computation and Natural Computation 2021, Espoo, Finland,

29th International Conference on DNA Computing and Molecular Programming, Sendai, Japan.

Published in: Cho, DJ., Kim, J. (eds) Unconventional Computation and Natural Computation. UCNC 2024.

Lecture Notes in Computer Science, vol 14776. Springer, Cham.

https://doi.org/10.1007/978-3-031-63742-1_14.

6.2 Artistic works

2024 — Director and animator for the 2D short animation “Aurinko and Selene” as part of the MA thesis.

Exhibited in the:

Lines of Descent exhibition in Helsinki Central Library Oodi, Helsinki, Finland,

Media Lab 30 Years exhibition, Aalto University, Espoo, Finland.

2023-2024 — Projection designer for the Rottien pyhimys musical.

46 shows performed across:

Musiikkiteatteri Kapsäkki, Helsinki, Finland,

Aleksanterin teatteri, Helsinki, Finland.

2023 — Projection designer for the Ihmisen jälkeen / Posthuman chamber opera.

3 shows performed in Dance House Helsinki, Helsinki, Finland.

2023 — Director and animator for “Takeoff” slam poetry performance. Collaboration with two other directors on a short animation visualizing a poem by Giovanni Baudonck using 3D procedural modeling. A part of the Urban Travel Machines project, co-funded by the Creative Europe Programme of the European Union.

Presented as an immersive slam poetry performance in the planetariums:

Prima Vista Literary Festival 2023, Tartu, Estonia,

Brussels Poetry Fest 2024, Brussels, Belgium,

Dome Dreaming Festival 2026, Stockholm, Sweden.

Official Selection in:

ZEBRA Poetry Film Festival 2023, Berlin, Germany,

Tokyo International Short Film Festival, Tokyo, Japan,

New York Tri-State International Film Festival, New York, USA,

Les Instants Vidéo, Marseille, France.

2022 — Director and animator for the 3D short animation “Dysmorphia”. A part of 3D animation minor in the MA studies.

Presented in:

Media Lab 30 Years 1999-2022 Animation Showreel,

Spring Demo Day 2022, Aalto Media Lab.

Official Selection in:

Fredrikstad Animation Festival 2022, Fredrikstad, Norway,

15th Annual Short Short Story Film Festival, Bethel CT, USA,

23. Animatricks International Animation Festival, Helsinki, Finland,

17th Anim!Arte - International Student Animation Festival of Brazil, Rio De Janeiro, Brazil,

Kinomural 2023, Wrocław Poland.

Grants:

2022 Department of Art and Media grant for travelling to the festival in Fredrikstad.

6.3 Software

2022 — Software for DNA visualization in the Natural Computation research group, Aalto University. MIT open source, publicly available at the repository: <https://github.com/dnaforge/xdsd>

7 Awards and honours

2023 — Open Call Winner for ”Takeoff” animation in the new media urban festival Kinomural 2023, Wrocław, Poland.

2021 —Recipient of the Warsaw University of Technology Rector’s scholarship for high academic performance for 2020/21 in the Computer Science degree. Graduated summa cum laude.